

Part D — Performance Analysis and Graphs

Case Study: The Commute Question

D1 — Relationship Interpretation

The calculated correlations are:

Relationship	r	Interpretation
Commute Time vs Final Score	-0.0593	Very weak negative linear relationship
Study Hours vs Final Score	0.6449	Moderate to strong positive linear relationship
Sleep Hours vs Final Score	0.1643	Weak positive linear relationship

Ranking: 1) Study Hours vs Final Score (0.6449), 2) Sleep Hours vs Final Score (0.1643), 3) Commute Time vs Final Score (-0.0593). Study hours have the strongest linear relationship with final score.

D2 — Correlation $\neq$ Causation

Correlation shows association, not proof that one variable causes another. Possible alternative explanations include: (1) students with longer commutes may have less time available for studying; (2) socioeconomic conditions, stress, or home environment may affect both commuting and academic performance. Longitudinal data, information about confounding variables, and an appropriate controlled or quasi-experimental study would help investigate causation.

D3 — Defining Long Commute

Q1 = 20.55 minutes, Q3 = 58.3 minutes, IQR = 37.75 minutes, and the upper IQR fence = 114.925 minutes. The selected threshold is Q3 = 58.3 minutes because it identifies the top 25% of students with the longest commutes while keeping enough students in both groups.

Threshold	Short	Long
Median = 33	130	129
Q3 = 58.3	194	65
Upper Fence = 114.925	244	15

Using Q3 = 58.3 minutes: Short commute group = 194 students; Long commute group = 65 students.

Measure	Short Commute	Long Commute
Mean Final Score	79.79	78.02
Median Final Score	79.30	78.15
Standard Deviation	9.57	10.04

The difference in mean final scores is $79.79 - 78.02 = 1.77$ points. This is small compared with the within-group standard deviations. Repeating the analysis with another random sample may produce a different gap; this variation is related to sampling error.

D4 — Choose the Right Graph

The following table uses separate, wrapped rows and a landscape-style layout within the page width to prevent text overlap.

Situation	Graph	Why?
Show the shape of the commute distribution	Histogram	Displays the frequency distribution of commute times across intervals.
Show that most students have low commute times with a long right tail	Histogram	Makes concentration at lower values and right skewness visible.

Situation	Graph	Why?
Compare commute times across five branches	Box Plot	Compares medians, spread, and possible outliers across branches.
Show whether study hours relate to final score	Scatter Plot	Shows direction, strength, and pattern between two numerical variables.
Show the most preferred transport modes	Bar Chart	Compares counts or frequencies of categorical transport modes.
Show satisfaction levels from 1 to 5	Bar Chart	Satisfaction levels are ordinal categories, so a bar chart is appropriate.

Final Summary

Study hours have the strongest positive linear relationship with final score. Commute time has an almost negligible linear correlation with final score. With $Q3 = 58.3$ minutes as the threshold, the mean score difference between the two commute groups is 1.77 points, which is small compared with the variation within the groups. These results describe associations and do not establish causation.